ReuseOps

Project Specifications — Reusable Items Platform

Live app: https://reuseops.lovable.app

Demo video: https://youtu.be/NQTnBzg5EzU

Tagline: Use. Return. Ready again.

Date: 12 October 2026 | Status: Working demo, verified end to end

1. Overview

ReuseOps is a mobile-first platform for tracking reusable items — cups, plates, containers, cutlery and shared event/campus items — across networks of venues. It keeps a persistent, auditable record of every item: where it is, what state it is in, and who is responsible for it. Items move through a controlled loop: issued → returned → collected → washed → inspected → ready again.

The Helsinki and Amsterdam workspaces in the app are clearly labelled as proposed demo settings, not confirmed partnerships. Demo data is never presented as real-world impact.

2. Core Principles

- Return-first loop: a small stock of reusables serves many people; every recorded return keeps counts correct and impact provable.
- Last confirmed checkpoint: an item's location is always the last place a person confirmed it — never a guess.
- Staff-confirmed handover: a QR scan never proves physical handover. Borrow and return requests only change item state when staff confirm.
- No fabricated impact: impact estimates appear only when an organiser supplies an assumption with a source; demo data is excluded.
- Append-only history: every action is a permanent transaction; duplicate scans cannot inflate counts (idempotent request IDs).
- Compliance honesty: the app records checks; it does not certify hygiene or safety compliance.

3. User Roles

Role	Capabilities
Borrower	Browse public areas, scan/enter codes, request to borrow or return, view own loans and due dates, cancel own pending requests.
Venue staff	All borrower abilities, plus confirm/decline custody requests at assigned areas (confirm issued / confirm received / received damaged).
Operator	Run collection, washing, inspection, cleaning and maintenance tasks across the network.
Organiser	Full administration: manage items, categories, locations/areas, staff assignments, impact assumptions, transfers, demo reset.

4. Item Lifecycle & States

Every item has a single current state. Transitions are validated server-side against the item's category rules (washing required, inspection required, maintenance allowed, etc.). Invalid actions are blocked.

State	Meaning	Next actions
ready	Available to borrow at its location	Issue / borrow request
issued / borrowed	Out with a borrower, due date tracked	Return request (any compatible venue or return station)
returned	Dropped off, awaiting collection	Collect (creates washing or inspection path)
collected	Picked up by an operator	Send to washing hub
washing / cleaning	Being washed or cleaned	Complete wash → inspection
awaiting inspection	Waiting for a pass/fail inspection	Pass → ready; fail → maintenance or quarantine
maintenance	Being repaired	Finish → inspection
quarantined / retired	Removed from circulation	Organiser decision

Category flags control the workflow per item type: washing_required, cleaning_allowed, inspection_required, maintenance_allowed, return_deadline_hours, compatible locations, max uses, deposit amount.

5. Feature Specification

5.1 Public & anonymous use

- Near me (/near): map and list of shared public areas with address search, device geolocation (only on demand, never stored), and filters: item group, borrow/return, open now, category compatibility. Demo areas shown separately as not real locations.
- Scan (/qr): camera scanning and photo upload of QR codes, plus manual code entry. External URLs are refused; unknown codes get a helpful message; codes existing in multiple pools offer a choice.
- Item passport (/i/<id>): public item page — status, location, borrow/return request buttons. Sign-in redirect remembers the page.
- Area page (/a/<id>): public area page with opening hours, accepted categories, availability, directions link — plus the staff confirmation queue for signed-in staff.

5.2 Signed-in use

- My loans: active loans with due dates, pending requests with cancel.
- Overview dashboard: live counts (ready, on loan, in wash loop, attention needed), overdue loans, open tasks, latest activity — all from saved transactions only.
- Scan & actions: location picker + code entry; only actions valid for the item's state, category and the user's role are offered.
- Inventory: every item with status, owner venue, last confirmed checkpoint; organisers add items in bulk by prefix + count.
- Operations: collection, washing, inspection, cleaning and maintenance tasks created automatically as items move; assignable to operators.
- Network: per-venue stock and transfer history; ownership stays with the issuing venue.
- Impact & costs: counts from recorded transactions only; estimates only where assumptions have values and sources; CSV export; demo data excluded.

- Kits: multi-item kits (e.g. lunch kit) with partial-return tracking and missing-item alerts.
- Deposits: deposit amounts recorded per category and tracked on issue/return.
- QR labels: printable labels encoding the public app URL + a stable public ID — no personal data.

6. Shared Areas & Custody Requests

Locations can be published as shared public areas with map coordinates, timezone, per-day opening hours, area type, accessibility notes, operator contact, return capacity, and flags for whether borrowing and/or returning is allowed and which categories are accepted.

Borrow/return flow: the requester scans or opens an item/area page and submits a request (idempotent via a client request ID). The request stays pending until staff assigned to that area confirm or decline it. Only on confirmation does the item change state and a transaction get recorded. One pending request per item is enforced.

7. Technical Architecture

Layer	Technology
Framework	TanStack Start v1 (React 19, SSR), Vite 7, deployed to edge runtime
Styling	Tailwind CSS v4, semantic design tokens, Fraunces + Source Sans 3, dark mode, reduced-motion support
Backend	Lovable Cloud: PostgreSQL database, authentication, row-level security, server functions (createServerFn)
Auth	Email/password with role selection at onboarding; roles stored in a protected user_roles table
Maps	Leaflet + OpenStreetMap (browser-only), Nominatim address search
QR	jsQR for camera/photo scanning; qrcode library for printable labels

8. Data Model

Table	Purpose
workspaces	Separate pools (e.g. Helsinki, Amsterdam); is_demo flag drives demo labelling and exclusion from impact.
user_roles	Membership + role per workspace (borrower / staff / operator / organiser).
locations	Venues, return stations, washing hubs, maintenance points; public_id, coordinates, hours, borrow/return flags, accepted categories.
item_categories	Workflow rules per category (washing, inspection, maintenance, deadlines, deposits, max uses).
items	Individual items: status, owner + current location, last confirmed checkpoint, responsible user, use count, stable public_id.
kits	Multi-item kits with component tracking for partial returns.
loans	Open loans (one per item), due dates from category deadlines.
transactions	Append-only audit log of every confirmed action; unique request IDs prevent duplicates.
tasks	Collection / washing / inspection / cleaning / maintenance work, assignable to operators.
inspections	Pass/fail inspection records.

Table	Purpose
transfers	Network transfers between venues; ownership stays with the issuer.
deposit_records	Deposit amounts held and released.
impact_assumptions	Organiser-supplied values + sources that power impact estimates.
area_staff	Staff assignment to public areas for custody confirmation.
custody_requests	Pending/confirmed/declined borrow & return requests; one pending per item.

9. Security & Access Control

- Row-level security on every table; anonymous users get no direct table access.
- Public pages read only through security-definer functions (public_areas, public_area, public_item, resolve_code).
- All writes go through server functions that validate role, state transition and category rules atomically.
- Demo workspaces are publicly readable; operational data is restricted to members by role.
- Borrowers see only their own loans and requests; staff see queues only for their assigned areas.

10. Verified Demo Journeys

- Cup issued at Venue A → returned at Venue B → collected → washed → inspected (pass) → ready again.
- Lunch kit partial return with missing-item alert.
- Shared item returned damaged → maintenance → inspection → available.
- Anonymous browsing of map/scan; signed-in borrow request → staff confirmation → loan with due date.

11. Demo Content

Helsinki and Amsterdam demo workspaces (labelled as proposed demo settings): Venue A and Venue B cafés, three return stations, a washing hub, a maintenance point; items CUP-001..008, PLATE-001..006, lunch kit KIT-001 (BOX-001, CUT-001, PLATE-K01), shared items SHARED-001/002. The demo can be reset by an organiser at any time. An empty pilot workspace can be created for real-world trials.

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